

1. Import Required libraries

```
In [152... import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [148... sns.set(style="whitegrid")
```

2. Load The Dataset

```
In [157... df = pd.read_csv("cleaned_train.csv")
```

3. Initial Data Exploration

```
In [104... df.info() # Check data types and null
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 10 columns):
 #   Column          Non-Null Count  Dtype
---  -
 0   PassengerId     891 non-null    int64
 1   Survived        891 non-null    int64
 2   Pclass          891 non-null    int64
 3   Sex             891 non-null    object
 4   Age             891 non-null    float64
 5   SibSp           891 non-null    int64
 6   Parch           891 non-null    int64
 7   Fare            891 non-null    float64
 8   Embarked        891 non-null    object
 9   Title           891 non-null    object
dtypes: float64(2), int64(5), object(3)
memory usage: 69.7+ KB
```

```
In [105... df.describe() # Summary statistics
```

```
Out[105...
```

	PassengerId	Survived	Pclass	Age	SibSp	Parch	F
count	891.000000	891.000000	891.000000	891.000000	891.000000	891.000000	891.000
mean	446.000000	0.383838	2.308642	29.361582	0.523008	0.381594	32.204
std	257.353842	0.486592	0.836071	13.019697	1.102743	0.806057	49.693
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000
25%	223.500000	0.000000	2.000000	22.000000	0.000000	0.000000	7.910
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454
75%	668.500000	1.000000	3.000000	35.000000	1.000000	0.000000	31.000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329

```
In [106... df.isnull().sum() # Missing values
```

```
Out[106... PassengerId    0
Survived      0
Pclass        0
Sex           0
Age           0
SibSp         0
Parch         0
Fare          0
Embarked      0
Title         0
dtype: int64
```

```
In [107... df.head() # Preview first few rows
```

```
Out[107...
```

	PassengerId	Survived	Pclass	Sex	Age	SibSp	Parch	Fare	Embarked	Title
0	1	0	3	male	22.0	1	0	7.2500	S	Mr
1	2	1	1	female	38.0	1	0	71.2833	C	Mrs
2	3	1	3	female	26.0	0	0	7.9250	S	Miss
3	4	1	1	female	35.0	1	0	53.1000	S	Mrs
4	5	0	3	male	35.0	0	0	8.0500	S	Mr

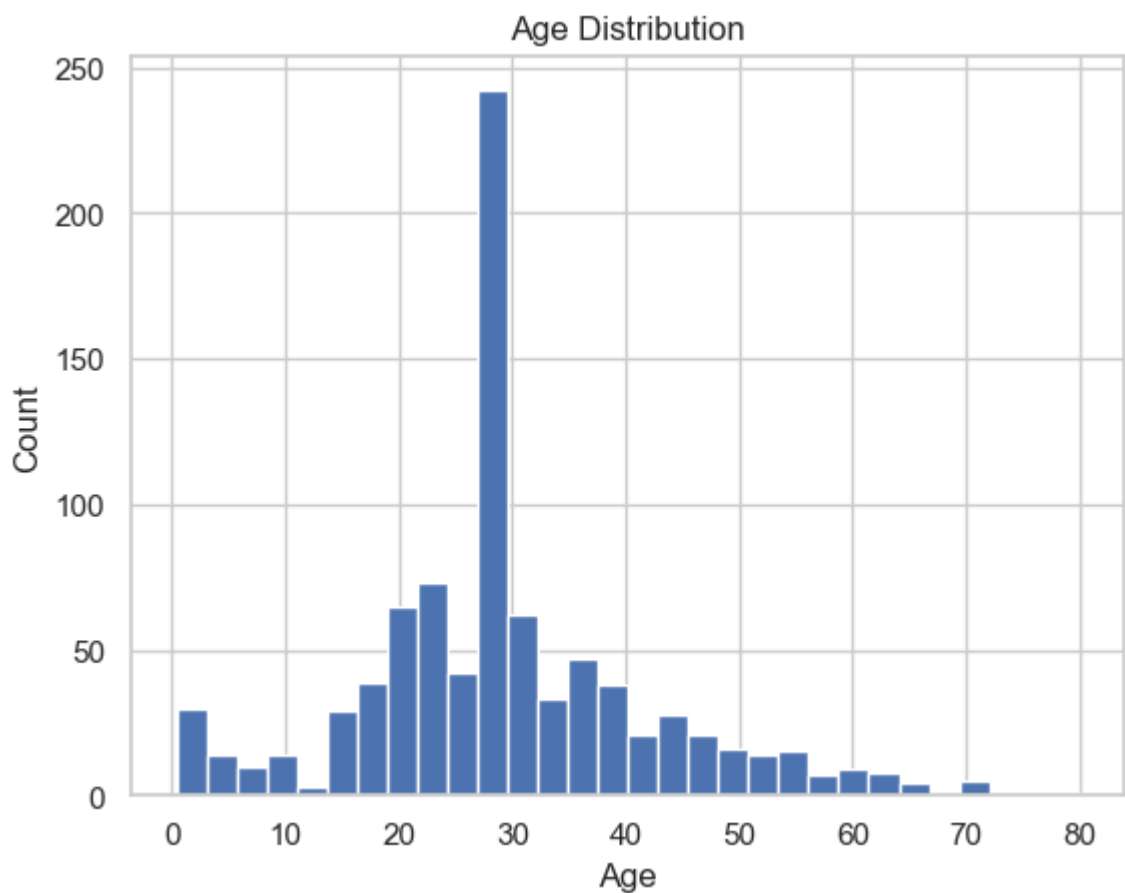
4. Value counts for categorical features

```
In [169... df['Sex'].value_counts()
df['Embarked'].value_counts()
df['Pclass'].value_counts()
df['Title'].value_counts()
```

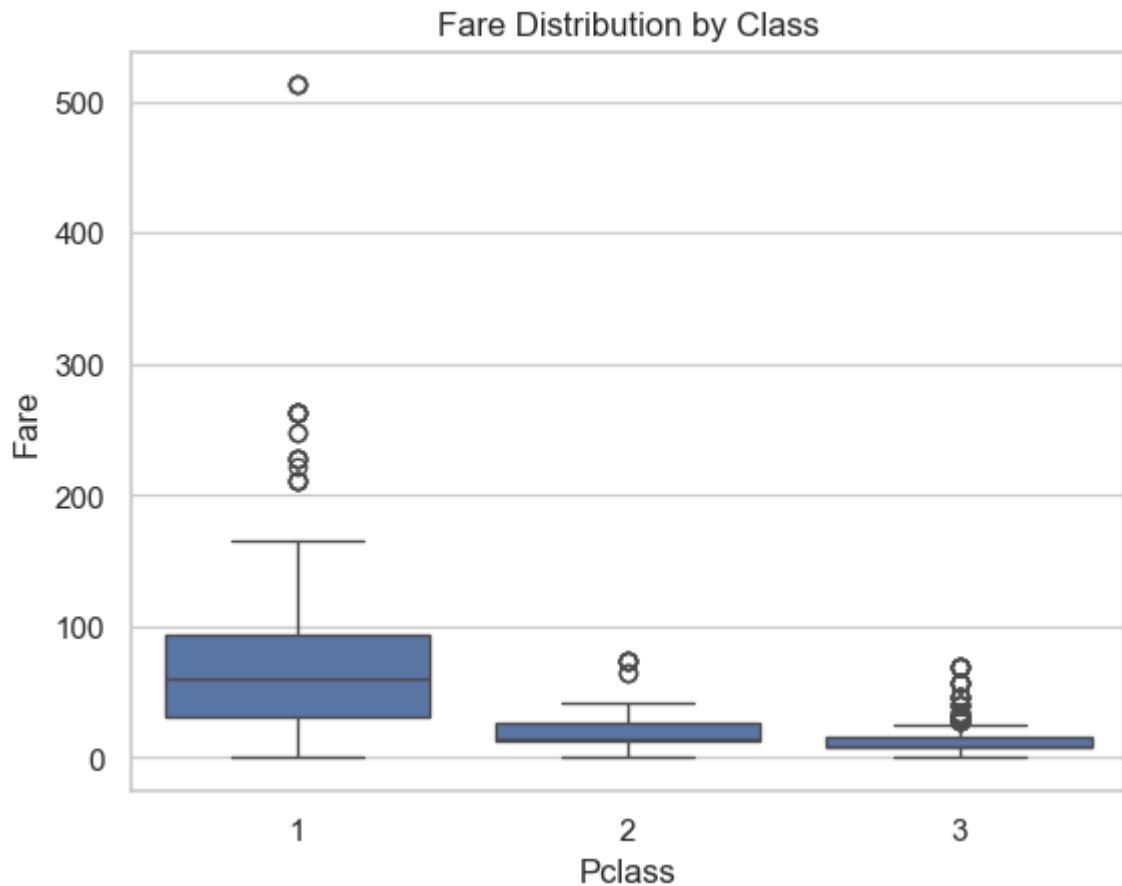
```
Out[169... Title
Mr          517
Miss        182
Mrs         125
Master       40
Dr           7
Rev          6
Mlle         2
Major        2
Col          2
Countess     1
Capt        1
Ms           1
Sir          1
Lady         1
Mme          1
Don          1
Jonkheer     1
Name: count, dtype: int64
```

5. *Univariate Analysis (Single Column)*

```
In [110... # a. Histogram
df['Age'].hist(bins=30)
plt.title("Age Distribution")
plt.xlabel("Age")
plt.ylabel("Count")
plt.show()
```



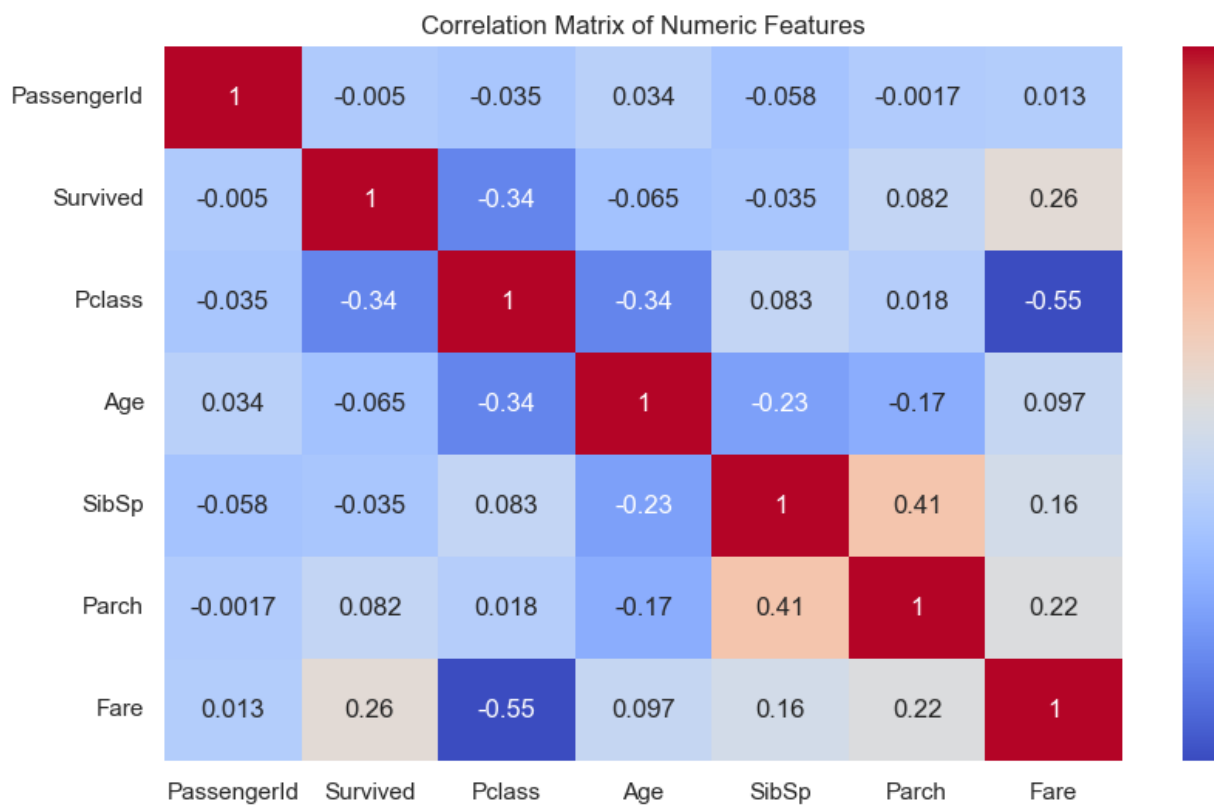
```
In [111... # b. Boxplot
sns.boxplot(x='Pclass', y='Fare', data=df)
plt.title("Fare Distribution by Class")
plt.show()
```



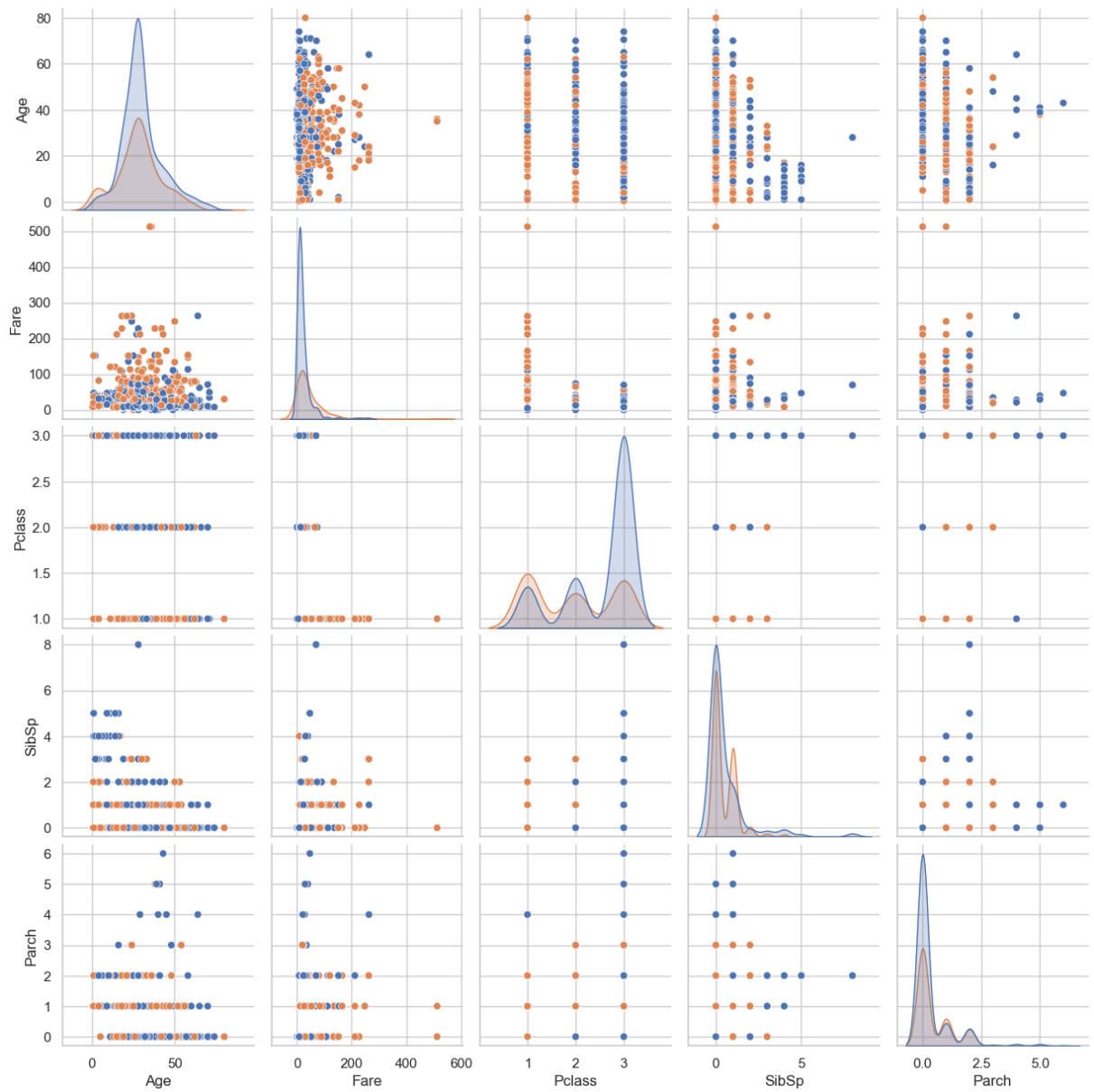
6. Bivariate/Multivariate Analysis

```
In [113... # a. Correlation Heatmap
# Only select numeric columns for correlation
numeric_df = df.select_dtypes(include=['int64', 'float64'])

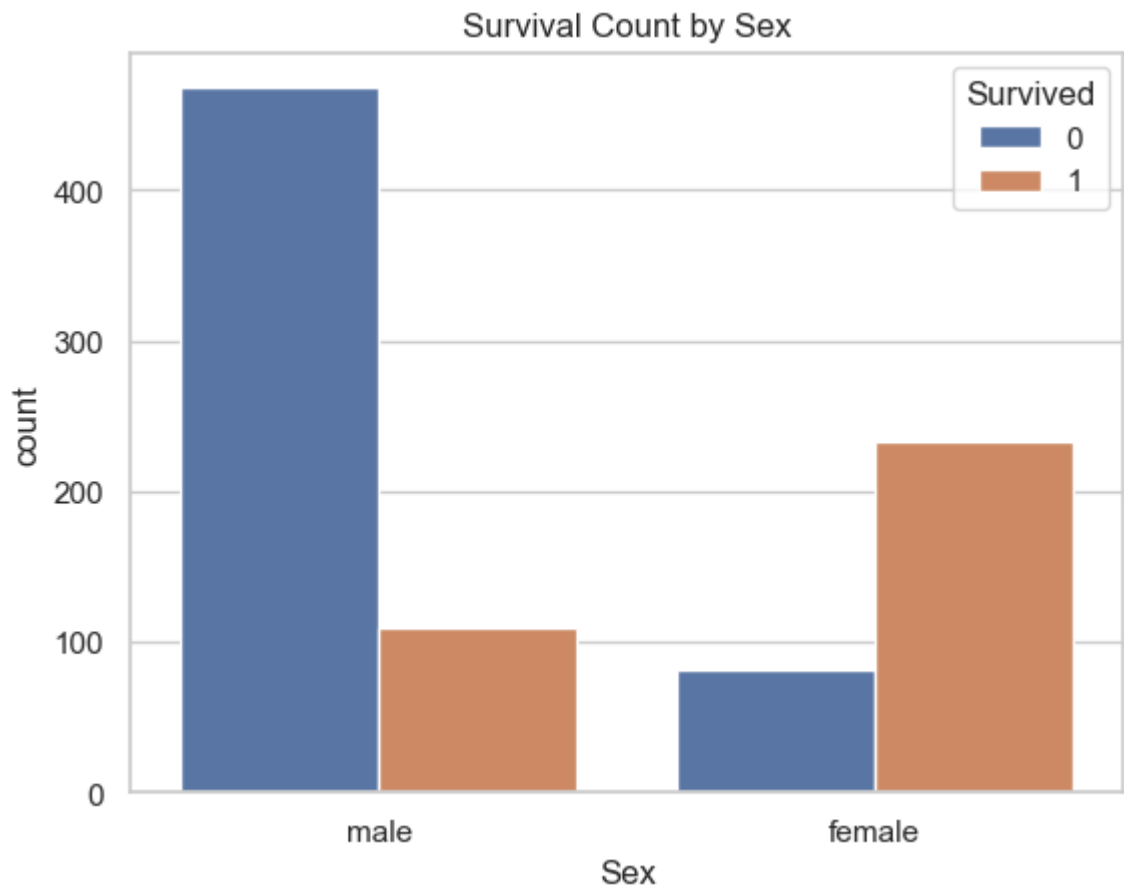
# Plot correlation heatmap
plt.figure(figsize=(10, 6))
sns.heatmap(numeric_df.corr(), annot=True, cmap="coolwarm")
plt.title("Correlation Matrix of Numeric Features")
plt.show()
```



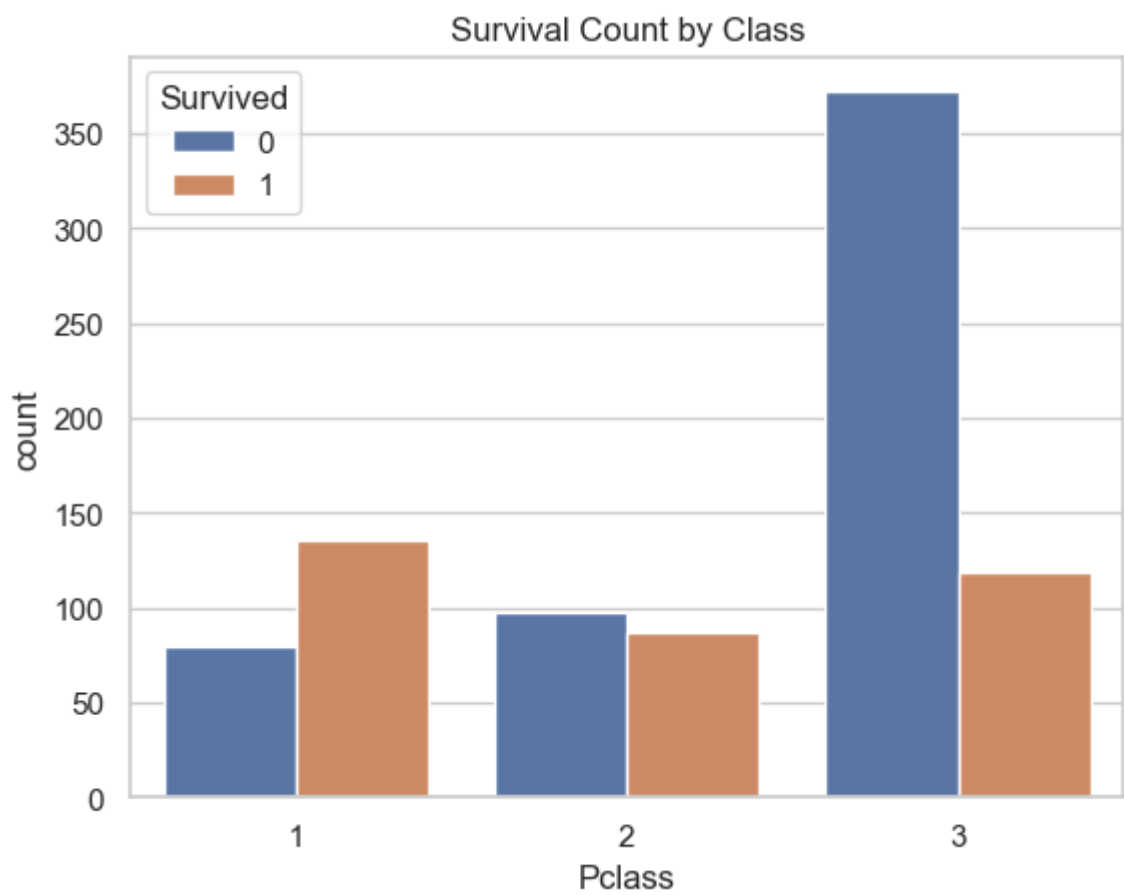
```
In [114]: # b. Pairplot
sns.pairplot(df, hue='Survived', vars=['Age', 'Fare', 'Pclass', 'SibSp',
plt.show()
```



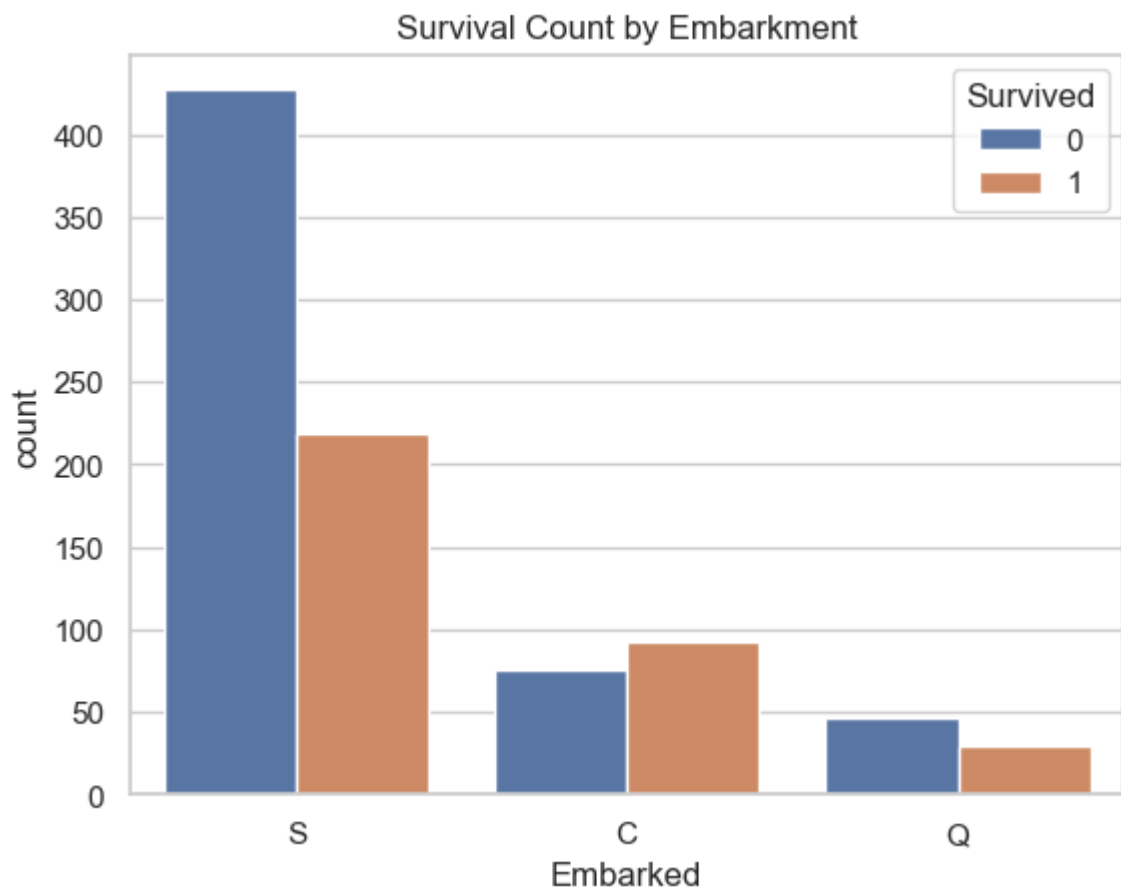
```
In [140... # c. Countplots (Categorical vs Survived)
# a)
sns.countplot(x='Sex', hue='Survived', data=df)
plt.title("Survival Count by Sex")
plt.show()
```



```
In [142... # b )  
sns.countplot(x='Pclass', hue='Survived', data=df)  
plt.title("Survival Count by Class")  
plt.show()
```



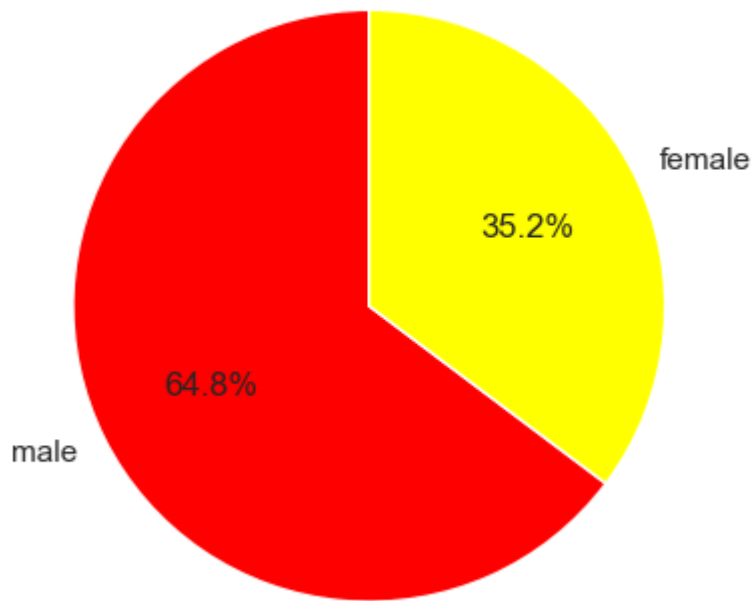
```
In [144... # c )
sns.countplot(x='Embarked', hue='Survived', data=df)
plt.title("Survival Count by Embarkment")
plt.show()
```



7. Fancy Visuals

```
In [178... df['Sex'].value_counts().plot.pie(autopct='%1.1f%%', startangle=90, color:
plt.title("Gender Distribution")
plt.ylabel('')
plt.show()
```


Gender Distribution



8. *Final Summary*

Gender and Survival

- Females had a significantly higher survival rate compared to males.
- This supports the "women and children first" evacuation protocol.

Passenger Class (Pclass)

- 1st class passengers had the highest survival rate.
- 3rd class passengers faced the lowest survival rate.
- Socioeconomic status clearly impacted access to lifeboats and safety.

Embarkation Port

- Passengers from Cherbourg (C) had a higher chance of survival.
- Majority embarked from Southampton (S), where survival was relatively lower.

Age Distribution

- Most passengers were between 20–40 years old.
- Children under 10 had a better survival rate than older age groups.

Fare Insights

- Survivors generally paid higher fares.
- High-fare tickets were linked to better accommodation and quicker access to lifeboats.

Family Features (SibSp, Parch)

- Passengers with 1 or 2 family members onboard had better survival rates.
- Being alone or having too many dependents reduced chances of survival.

Title Feature (Mr, Miss, Mrs, etc.)

- Titles added valuable insight:
- Miss and Mrs had higher survival rates.
- Mr had the lowest, aligning with male survival statistics.

Correlation Findings:

- Survived was positively correlated with:
- `Fare`, `Sex` (when encoded), and `Pclass` (inversely).
- `SibSp` and `Parch` showed minor correlations.
- Categorical variables like `Sex` and `Title` were highly predictive after encoding.

In []: